

Pre-feasibility Study for the Establishment of Wholesale Vegetable and Fruit Market in Al-Tafilah



2017



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1. Executive summary

The project aims to establish a vegetable and fruit market in Tafileh Governorate, which is located in the southern part of Jordan. The governorate lacks a high standard central vegetable and fruits market, hence, arise the idea of the establishment of a market targeting the people of the governorate mainly and other surrounding governorates of southern Jordan.

By implementing the project, the level of services in the region will be improved, in addition to the provision of job opportunities for the citizens.

The table below shows the financial indicators of the project:

Table1 project indicators

indicator	Value
The Overall investment Cost (JD)	349,035
Internal Rate of Return	%18.16
Payback Period (Years)	7
Net Present Value (JD)	107.098
Jobs Opportunities provided by the project	4
Expected Annual revenues (for 10 Years)	1,432,985
Total Operating Expenses (for 10 years)	189,154
Gross Profit (for 10 years)	274.316

Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

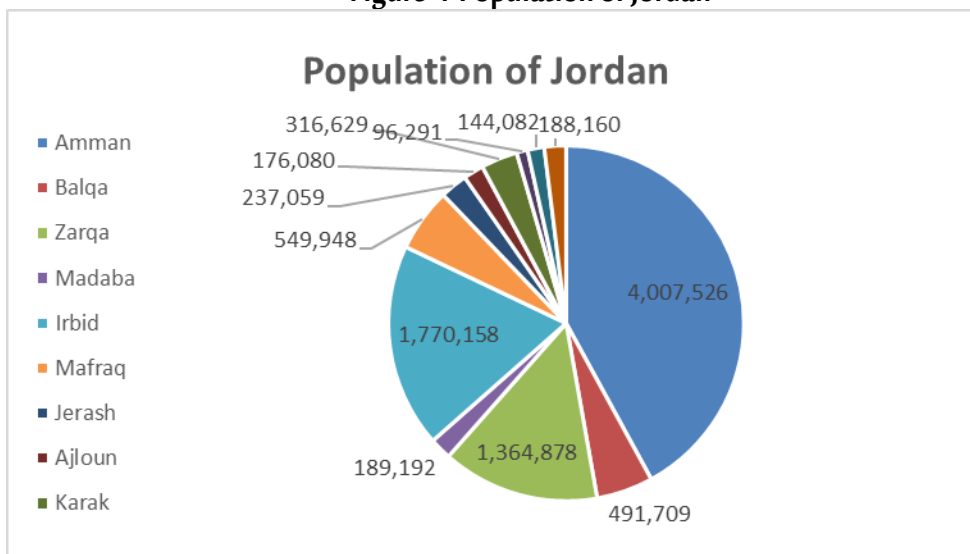
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafraq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table2 Population Distribution according to Age Group

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

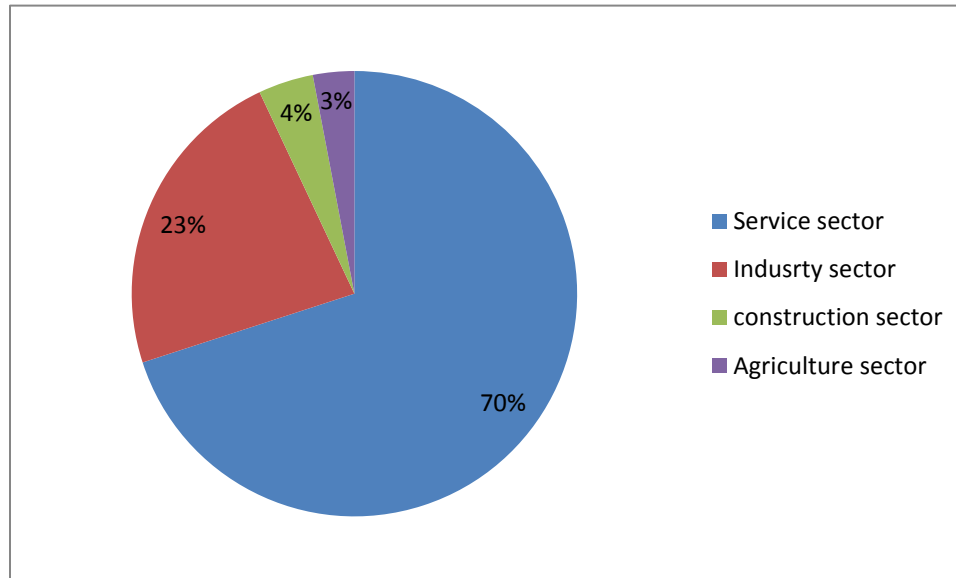
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure2 Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and comprising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and

medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

furthermore, in it endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of

capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

Investment Climate in Al-Tafeilh:

Under the provisions of fifth article of the investment law No. 30 of 2014 and Income tax regulations in the less developed areas, the reduction of the income tax system on the investment projects are clarified in the table below:

Table 1 Income Tax and Exemptions

Within the Development Zone or Free Zone	Less Developed Areas
Income Tax is 5%	Income Tax is 20% As per Investment Law categorization, Irbid is classified in Class (b) where investors enjoys 80% exemption on Income tax for 20 years, where net income tax after reduction should not be less than 5%.

2. Market Analysis

Project Description

The idea of the proposed project is to establish a central wholesale market for vegetables and fruits in the governorate of Tafileh adopting a BOT system. This means that the land will be leased, developed and managed by an investor, with granting part of project profits to the government. After ten years from starting the project, the government will regain the land with its properties.

The investor will divide the land and lease it to the vegetables and fruits traders to sell their products at a competitive price in the market.

Project Objectives

Project objectives are summarized below:

- Achieving a profitable return to project investor .
- Providing a central specialized wholesale market for vegetables and fruits in AlTafilah.
- Providing vegetables and fruits at competitive prices.
- Providing job opportunities for the residents of AlTafilah Governorate.

Proposed Service

The proposed project will provide a central specialized wholesale market for selling vegetables and fruits in AlTafilah governorate in order to serve its residents.

Targeted Segments

The proposed project will target the following segments:

- Mainly the residents of AlTafilah Governorate.
- Visitors and residents of the Southern governorates.

Market Size

Preliminary data released by the Department of Statistics in 2013 have shown the dominance of planted products in the pattern of food consumption per capita in Jordan, where vegetable products accounted almost 87% of the daily share of the total calories per capita.

The total production of vegetables in Jordan for the year 2015 reached 2,047,437 tons. The total fruit production of Jordan amounted to 864,171 tons. The average consumption per capita of vegetables amounted to 110.6 kg / year, while the consumption rate of fruits was 61.8 kg. / Year.

Tafilah governorate:

The following are the indicators of the Agricultural Sector in Tafilah Governorate compared to the Kingdom:

Table 3 Agricultural Sector Indicators of AlTafilah Governorate compared to the Kingdom:

Indicator	Tafilah	Kingdom	%
Planted field with Crops (donum)	76490	705594	0.108
Planted field with Fruits (donum)	29190	1632929	0.018
Planted field with Vegetables (donum)	3092.5	520760	0.006
Total (donum)	108772.5	2859283	0.038
The number of agricultural cooperatives *	16	330	0.05

Source: Department of statistics, 2015.

* Source: Jordan cooperative enterprise, 2015.

Competitors Analysis

The Vegetable and Fruit wholesale market will be the first of its kind in AlTafilah Governorate, as there are no direct competitors in the governorate except small scattered vegetable and fruit shops.

Project Capacity

An area of 2000 ^m₂ was allocated for the project, divided into 50 shops with a total area of 25 m² per shop, which equals to 1,250 Sq.m of the total land area. All shops are expected to be leased in the first year of the project.

Price Analysis

Based on the Market rental prices in AlTafilah Governorate, the proposed price for land lease is 20,000 JD with the assumption that the average rental price of each shop is 2,500 Jordan dinars annually.

Projected Revenues

After analyzing market size and determining project capacity, the following table shows the projected revenues of the project over the years (2018-2027):

Table 4 Projected Revenues for 10 years

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Shop	2500.0	2575.0	2652.2	2731.8	2813.7	2898.1	2985.1	3074.6	3166.9	3261.9
Rentals	0	0	5	2	7	9	3	8	3	3
Total Revenue	125000	128750	132613	136591	140689	144909	149257	153734	158346	163097

3. Technical analysis

Required Human Resources

The following table shows the labor for the project:

Table 5 required labor for the project

JobTitle	No.
Indirect Employees	
Project Manager	1
Accountant	1
Cleaner	2
Total	4

Job DescriptionS:

- **Project manager:** Suggesting the general policy of the central wholesale market, preparing special plans and follow-up their implementation, performing technical and administrative supervision, and developing procedures and ensuring the implementation of occupational safety and health requirements.
- **Accountant:** Follow-up on all financial and auditing procedures, and submit annual reports of project financial situation.
- **Cleaner:** Maintaining the cleanliness of the central wholesale market.

Site Analysis

The project will be built on an area of 2000 m² so that it will be divided into shops, with a space reserved for cars and goods cars. It is proposed that the project be implemented in the center of AlTafilah - Municipality Street.

Technical Requirements Analysis

Land and Construction

The following table shows project required land:

Table6 Project required Land

Land	Area Sq.m
The land required	2,000

The required construction works for the project are shown as follows:

Table 7 Construction work required

Construction Work	Area (Sq.m)
Hanger	1,250
Paving	2,000

Machinery and Equipment

The following table provides the machinery and equipment needed for the project:

Table 8 Machinery and Equipment

Equipments	Quantity
Computer	2

Furniture

The nature of the project requires a very simple furniture, such as carpets.

4. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 12.05% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Income tax on the net project income is 5 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 2 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 3 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 4 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 5 Expenses Assumptions

Expenses Assumptions	
Utilities Expenses from Total Revenues	10%
Maintenance Expenses from Total Revenues	1%
Depreciation Expenses from Total Revenues	1%
Insurance Expenses from Total Assets Value	0.75%
Marketing Expenses from Total Revenues	0.05%

Table 6 Income Tax Assumptions

Income Tax Assumptions	Value
(1) Average Income Tax in Jordan	20%
(2) Income Tax Deduction	0%
Income Tax after Deduction	20%
(3) Compulsary Reserve Percentage	10%
Other Assumptions	Value
(1) Annual Monthly Salaries have been calculated after multiplying monthly salaries by:	16

Risk Premium	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	9.48%
(3) Market Risk Premium	10.12%
Income Tax Rate	%20
(4) Beta	1
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table7 : Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premium	%10.12
Income Tax Rate	%20
Bets	1.00
Equity	139,614
Loans Value	209,421
Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%7.58
Loans Value	209,421
Loans Percentage	%60

Equity	
Cost of Equity	%16.6
Equity Value	139,614
Equity Percentage	%40
Gross	
Project value	349,035
Percentage of project	%100
weighted average cost of capital	%11.20

Capital Expenditures

The estimated cost of the project is 349,035 JD and it covers construction works costs, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 8 Construction Work Capital Expenditure

Construction Works	Cost (JOD per Squared)	Area	Gross
Hunger	100	2,000	200,000
Paving	25	3,000	75,000
Other	0	-	20,000
Gross total			295,000

Table 9 machinery and equipments Work Capital Expenditure

Machinery and equipments	No.	Cost (JOD per	Gross cost
Computer	2	350	700
Total			700

Table 10 Furniture and furnishings Work Capital Expenditure

Furniture	No.	Cost (JOD per unit)	Gross cost
Furniture	3	500	1,500
Total			1,500

Table : 11 Pre Operating Expenses

Consist of a set of expenses that are expended before the start of the project and before achieving any revenue. Government fees, licenses, marketing expenses, training and transportation expenses are estimated as follows:

Pre-operating expenses	Estimated cost (JD)
Governmental fees	1,000
Transportation	250
Marketing	500
Training	1,750

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table12 Working Capital

Working Capital	No. of	%	Estimated Cost
Operating expenses	3	%25	4,125
General and administrative expenses	3	%25	6,614
Marketing expenses	3	%25	16
Indirect Salaries	3	%25	7,600
Total			18,355

Table 13 Expenses Summary

Capital expenditure	Cost (JD)	Contingency	Cost (JD)	%
Construction works	295,000	10%	324,500	%93.0
Machinery & Equipment	700	10%	770	%0.2
Furniture	1,500	10%	1,650	%0.5
Vehicles	1,750	10%	1,925	%0.6
Pre-Operating Expenses	18,355	10%	20,190	%5.8
Working capital	317,305		349,035	100%

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (349,035) JD, where 60% of the project will be financed through loans (209,241) JD, and the remaining (40%) through (Equity) with (139,614) JD. The following table summarizes the general structure of the required financing:

Table 14 Sources of Funding

Funding Sources	Amount	Percentage
Equity (self financing)	139,614	%40
Loans	209,421	%60
Total	349,035	%100
Use of Fund:	المبلغ	النسبة
Capital Expenditures	326,920	%93.66
Pre-Operating Expenses	1,925	%0.55
Working Capital	20,190	%5.78
Total	349,035	%100.00

Operating Expenses

The following table summarizes the expected operating expenses for the proposed project (2018-2027):

Table 15 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Utilities from Revenues	12,500	12,875	13,261	13,659	14,069	14,491	14,926	15,373	15,835	16,310
Maintenance from Revenues	1,250	1,288	1,326	1,366	1,407	1,449	1,493	1,537	1,583	1,631
Disposables from Revenues	1,250	1,288	1,326	1,366	1,407	1,449	1,493	1,537	1,583	1,631
Others	1,500	1,545	1,591	1,639	1,688	1,739	1,791	1,845	1,900	1,957
Total	16,500	16,995	17,505	18,030	18,571	19,128	19,702	20,293	20,902	21,529

General and Administrative Expenses

The following table summarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 16 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	600	618	637	656	675	696	716	738	760	783
Hospitality	250	258	265	273	281	290	299	307	317	326
Stationary and Printing	250	258	265	273	281	290	299	307	317	326
Transportation	500	515	530	546	563	580	597	615	633	652
Rent.	30,000	30,900	21,218	21,855	22,510	23,185	23,881	24,597	25,335	26,095
Insurance	2,452	2,525	2,601	2,679	2,760	2,842	2,928	3,016	3,106	3,199
Miscellaneous	2,405	2,477	2,552	2,628	2,707	2,788	2,872	2,958	3,047	3,138
Total	26,457	27,251	28,068	28,910	29,778	30,671	31,591	32,539	33,515	34,521

Marketing expenses

The following table shows the expected marketing expenses of the proposed project. Marketing expenses account for 0.01% of revenues based on the team's estimate:

Table 17 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total	63	64	66	68	70	72	75	77	79	82

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table9 number of employees and their wages (Jordanian dinars)

Job title	No.	Monthly salary	Total monthly salaries	Total annual wages
Indirect employees				
Project manager	1	1,000	1,000	16,000
Accountant	1	400	400	6,400
Cleaning labours	2	250	500	8,000
Total	4		1,900	30,400

The table below shows the expected project wages during a period of ten years from the start of the project:

Table : 18 Expected Annual Salaries and Wges (2018-2027)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Project manager	16,000	16,656	17,339	18,050	18,790	19,560	20,362	21,197	22,066	22,971
Accountant	6,400	6,662	6,936	7,220	7,516	7,824	8,145	8,479	8,826	9,188
Cleaning labours	8,000	8,328	8,669	9,025	9,395	9,780	10,181	10,599	11,033	11,485
Total	30,400	31,646	32,944	34,295	35,701	37,164	38,688	40,274	41,926	43,645

Depreciations

The following tables illustrate cost of Capex and Depreciation rates:

Table 19 Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Construction Works	324,500	%5.0	%0.0
Machinaries	770	%10.0	%2.0
Furniture	1,650	%10.0	%5.0
Vehicles	0	%10.0	%0.0
Gross total	326,920		

Table 20 Consumption and annual increases for construction

Capex, Annual Additions and Depreciations	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	326,920	327,018	327,120	327,227	327,339	327,456	327,578	327,706	327,840	327,980
Total Depreciation Expenses	16,467	16,477	16,487	16,498	16,509	16,521	16,533	16,546	16,559	16,573
Total Accumulated Depreciation	16,467	32,944	49,431	65,929	82,437	98,958	115,491	132,036	148,595	165,168
Total Additions	0	98	102	107	112	117	122	128	134	140
Total Net Book Values	310,453	294,074	277,689	261,299	244,902	228,498	212,087	195,670	179,244	162,811

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table : 21 Loan Details

Loan Value	209,421
Annual Interest Rate	9.48%
Loans Period	5
Loan Starts at year:	2018
Annual Payment	54,513
Number of Payments	5

The following table shows the loan payments

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2018				209,421
2019	54,513	19,853	34,660	174,762
2020	54,513	16,567	37,945	136,816
2021	54,513	12,970	41,543	95,273
2022	54,513	9,032	45,481	49,793
2023	54,513	4,720	49,793	0

Income Statement

All financial information and analysis of the financial expenses of the project shall be the estimated income statement for the years (2018-2027) as shown in the following table:

Table 22 Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	125,000	128,750	132,613	136,591	140,689	144,909	149,257	153,734	158,346	163,097
Gross Operating Revenues	125,000	128,750	132,613	136,591	140,689	144,909	149,257	153,734	158,346	163,097
Operating Expenses	-16,500	-16,995	-17,505	-18,030	-18,571	-19,128	-19,702	-20,293	-20,902	-21,529
Gross Operating Profit	108,500	111,755	115,108	118,561	122,118	125,781	129,555	133,441	137,445	141,568
<i>Gross Profit Percentage</i>	%87	%87	%87	%87	%87	%87	%87	%87	%87	%87
Salaries and Benefits (Indirect Staff)	-30,400	-31,646	-32,944	-34,295	-35,701	-37,164	-38,688	-40,274	-41,926	-43,645
General and Administrative Expenses	-26,457	-27,251	-28,068	-28,910	-29,778	-30,671	-31,591	-32,539	-33,515	-34,521
Marketing Expenses	-63	-64	-66	-68	-70	-72	-75	-77	-79	-82
Pre Operating Expenses	-1,925									
Gross Indirect Expenses	-58,845	-58,962	-61,079	-63,273	-65,549	-67,908	-70,354	-72,890	-75,520	-78,247
Income before Interest, Depreciation and Tax	49,655	52,793	54,029	55,288	56,569	57,873	59,201	60,551	61,925	63,321
Fixed Assets Depreciations	-16,467	-16,477	-16,487	-16,498	-16,509	-16,521	-16,533	-16,546	-16,559	-16,573
Income before Tax and Interests	33,188	36,317	37,542	38,790	40,060	41,353	42,668	44,006	45,366	46,748
Bank Interests	-19,853	-16,567	-12,970	-9,032	-4,720	0	0	0	0	0
Income Before Tax	13,335	19,749	24,572	29,758	35,340	41,353	42,668	44,006	45,366	46,748
Income Tax	-2,667	-3,950	-4,914	-5,952	-7,068	-8,271	-8,534	-8,801	-9,073	-9,350

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Profit	10,668	15,799	19,658	23,806	28,272	33,082	34,134	35,204	36,293	37,399
<i>Net Profit Percentage</i>	<i>%9</i>	<i>%12</i>	<i>%15</i>	<i>%17</i>	<i>%20</i>	<i>%23</i>	<i>%23</i>	<i>%23</i>	<i>%23</i>	<i>%23</i>
Compulsory Reserves	-1,067	-1,580	-1,966	-2,381	-2,827	-3,308	-3,413	-3,520	-3,629	-3,740
Retained Earnings	9,601	23,821	41,513	62,938	88,383	118,157	148,878	180,562	213,225	246,884

Expected Cashflows Statement

The table below summarizes project cash flows statement, where net cash flows are positive during all years except the first two years of project and that indicate to needs the loan as below:

Table 23 Expected cash flows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating Activities										
Net Profit	10,668	15,799	19,658	23,806	28,272	33,082	34,134	35,204	36,293	37,399
Bank Interests	19,853	16,567	12,970	9,032	4,720	0	0	0	0	0
Depreciation	16,467	16,477	16,487	16,498	16,509	16,521	16,533	16,546	16,559	16,573
Total Operating Cashflows before Additions to Working Capital	46,988	48,844	49,115	49,336	49,501	49,603	50,667	51,750	52,852	53,972
Inventory (Increase/Decrease)	2,750	83	85	88	90	93	96	99	101	105
Accounts Receivables (Increase/Decrease)	2,750	83	85	88	90	93	96	99	101	105

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Accounts Payables (Increase/Decrease)	49,738	48,926	49,200	49,424	49,591	49,696	50,763	51,849	52,953	54,076
Net Cashflows from Operating Activities										
Fixed Assets (Procurement)	-326,920	-98	-102	-107	-112	-117	-122	-128	-134	-140
Cashflows from Investments Activities	-326,920	-98	-102	-107	-112	-117	-122	-128	-134	-140
Net Cashflows from Investments Activities										
Capital	139,614									
Loan Amortization	-34,660	-37,945	-41,543	-45,481	-49,793	0	0	0	0	0
Bank Interest Rate	-19,853	-16,567	-12,970	-9,032	-4,720	0	0	0	0	0
Loans	209,421	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing Activities	294,523	-54,513	-54,513	-54,513	-54,513	0	0	0	0	0
Net (Increase/Decrease) in Cash	17,341	-5,685	-5,415	-5,196	-5,033	49,579	50,641	51,721	52,819	53,936
Cashflows at the Beginning of Period	0	17,341	11,656	6,241	1,045	-3,989	45,590	96,230	147,951	200,770
Cashflows at the End of Period	17,341	11,656	6,241	1,045	-3,989	45,590	96,230	147,951	200,770	254,706

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 24 Expected Balance Sheet

Description	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Assets										
Current Assets										
Cash	17,341	11,656	6,241	1,045	3,989-	45,590	96,230	147,951	200,770	254,706
Total current Assets	17,341	11,656	6,241	1,045	3,989-	45,590	96,230	147,951	200,770	254,706
Non Current Assets										
Fixed Assets (net)	310,453	294,074	277,689	261,299	244,902	228,498	212,087	195,670	179,244	162,811
Total Non Current	310,453	294,074	277,689	261,299	244,902	228,498	212,087	195,670	179,244	162,811
Total Assets	327,794	305,730	283,930	262,343	240,913	274,088	308,318	343,621	380,015	417,518
Liabilities										
Current Liabilities										
Paybles	2,750	2,833	2,917	3,005	3,095	3,188	3,284	3,382	3,484	3,588
Remaining amount of Loan	37,945	41,543	45,481	49,793	0	0	0	0	0	0
Total current Liabilities	40,695	44,375	48,398	52,798	3,095	3,188	3,284	3,382	3,484	3,588

Description	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Non Current Liabilities										
Long Terms Loans	136,816	95,273	49,793	0	0	0	0	0	0	0
Total Long Term Liabilities	136,816	95,273	49,793	0	0	0	0	0	0	0
Total Liabilities	177,512	139,649	98,191	52,798	3,095	3,188	3,284	3,382	3,484	3,588
Owners Equity										
Shareholders Contributions	139,614	139,614	139,614	139,614	139,614	139,614	139,614	139,614	139,614	139,614
Statutory Reseve	1,067	2,647	4,613	6,993	9,820	13,129	16,542	20,062	23,692	27,432
Retained Profits	9,601	23,821	41,513	62,938	88,383	118,157	148,878	180,562	213,225	246,884
Total Equity	150,282	166,082	185,739	209,546	237,817	270,900	305,034	340,238	376,531	413,930
Total Liabilities and Equity	327,794	305,730	283,930	262,343	240,913	274,088	308,318	343,621	380,015	417,518

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 25 Free Net Cash flows Table

Net Cashflows	Pre- Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final value
Net Free Cashflows	-349,035	51,663	48,828	49,097	49,317	49,479	49,579	50,641	51,721	52,819	53,936	779,252
Discount Factor	1.00	0.73	0.65	0.59	0.53	0.48	0.43	0.38	0.35	0.31	0.28	0.28
Net Present Value for Cash Flows	-349,035	37,574	31,936	28,878	26,086	23,536	21,208	19,481	17,893	16,433	15,090	218,019

Table below (42) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 26 Payback Period

Payback Period	Pre- Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	-349,035	51,663	48,828	49,097	49,317	49,479	49,579	50,641	51,721	52,819	53,936
Project Value Returned	349,035	297,372	248,544	199,446	150,130	100,651	51,072	431	-51,289	-104,109	-158,045
Payback Period (Year)	7	1	1	1	1	1	1	1	0	0	0

Table 27 Financial Analysis Results

indicator	Value
Weighted Average Cost of Capital (WACC)	%11.20
Net Present Value for Cashflows	107,098
Payback Period	7
Internal Rate of Return	%18.16

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Table 28 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%18.16	6	11.20%
Revenues declined by 10%	%14.42	8	11.20%
Operating Expenses Increased by 10%	%17.62	7	11.20%

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 107,098 Jordan Dinars considering that the project provides 4 Job Opportunities for the governorate resident